Pierre and Xavier cut a polygon

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A regular polygon with $n \geq 3$ sides of length $a > 1$ is drawn on the board, and the following game is played.

Pierre draws a diagonal (not a side) of the polygon, and Xavier erases one of the two open half-planes bounded by this diagonal. This goes on until only a triangle remains on the board ; let x, y and z be its side lengths. Xavier wins if the numbers $x^2 - 1$, $y^2 - 1$ and $z^2 - 1$ are again the side lengths of a non-degenerate triangle, and loses otherwise.

Determine, depending on n, which player has a winning strategy.